

MECH70034-UES January 2026

## MSc in Sustainable Energy Futures – Urban Energy Systems module

### Coursework project: Design an energy system for an urban area

Submission deadlines:

group report: 16:00 on Friday the 23<sup>rd</sup> of January 2026

individual report: 18:00 on Friday the 23<sup>rd</sup> of January 2026

#### 1 Background and context

As will be demonstrated during the lecture programme, space and water heating in buildings as well as electricity use accounts for a large share of the energy and carbon dioxide budget in cities around the world. Finding efficient solutions for meeting energy demands in cities is essential in climate change mitigation, as well as improving affordability and reducing dependencies on imported resources. Governments plan for regulations and standards to ensure energy efficiency, may set limits for emissions, mandate the use of renewable energy technologies, provide incentives for more sustainable alternatives, and/or ban certain technologies completely, in order to steer the decisions made by developers, home owners, and energy companies.

An example of this, currently under development by the UK government, is the Future Homes and Buildings Standards which have been under consultation in recent years and are expected to come in effect in England in late 2026 with a transition period to late 2027. Scotland is working on similar new building standards, including Passivhaus equivalent standards which might get implemented from 2028, and it has already phased out oil and gas boilers in new builds since 2024. This aims to set performance requirements at a level which ensures new homes and non-domestic buildings have high fabric standards, use low-carbon heating and are 'zero-carbon ready'. While we are not going to look at all the specifics of such proposals and the small print of this standard and other policies, the direction of travel is clear and in this coursework project you will be asked to explore sustainable urban energy systems and evaluate various options systematically.

Several studies have been undertaken on the use of heat pumps, combined heat and power (CHP) and district/community heating (DH/CH), distributed electricity generation, improved insulation and behaviour change, etc. However, they draw different conclusions on the potential for the type of technology and the choice of solution ultimately depends on the unique circumstances of an individual project. Energy can be imported (e.g., from the city's hinterland, elsewhere in the country or even from abroad) or generated locally within the city, and conversion technologies of different scales can be positioned inside individual buildings, within the urban area or placed outside which has an impact on distribution, transport, efficiencies, and ownership structures, among other things.

When deciding which solution to implement, a wide range of criteria has to be considered. Moreover, these criteria will be different for various stakeholders including residents, local government, energy suppliers, network operators, etc. Objectives can include minimising costs (but for whom?), minimising carbon emissions (but at what cost?) or maximising the efficiency of the overall system (but how to share the benefits of that between e.g. suppliers and network operators?).

During the lectures several methods will be introduced that can be used to explore different scenarios and evaluate various system interventions and policies, including optimisation, agent-based simulation,

building simulations, and value chain models. In this project the focus is on the use of optimisation tools to decide on the design of an urban energy system, though you are encouraged to refer to other methodologies in your report to explain how they could be used in further analysis.

## 2 Project description

In this project a new greenfield community development in Scotland is being planned and **your task is to develop an energy system design to provide space/water heating and power to the district**. The land use plan has been agreed and is outlined in Figure 1. The blueprints include plans for housing at various densities, a school, small hospital, a shopping centre, and an office building, as well as green space and roads. Some space is also to be left open for potential future development. Available energy demand data for the development and some suggested cost and related data are available in the accompanying Excel spreadsheet. Also consider the reference list below. You can look for additional or alternative data sources if and when needed, or choose scenarios in which this data has changed. It could be particularly interesting to explore how robust your solution is if energy prices go up further, or perhaps fall, and the same for capital costs for new technologies. Please document this clearly and cite all relevant sources.

First, you should try to establish a **baseline scenario** (which could be considered a no-mitigation pathway, which utilises gas boilers and grid electricity – though note that since 2025 Scotland does not actually allow gas boilers in new builds so it's really just a comparison point!), and afterwards you can explore **alternative designs** which move away from this baseline and describe how they impact the criteria selected. You should provide cost estimates (on an appropriate basis) and annual CO<sub>2</sub> emissions (on an operational rather than life-cycle basis) for your design. Please indicate what alternatives were studied, why those were chosen, and how they compare, using analytical methods including an optimisation model to evaluate them.

Your solution should include:

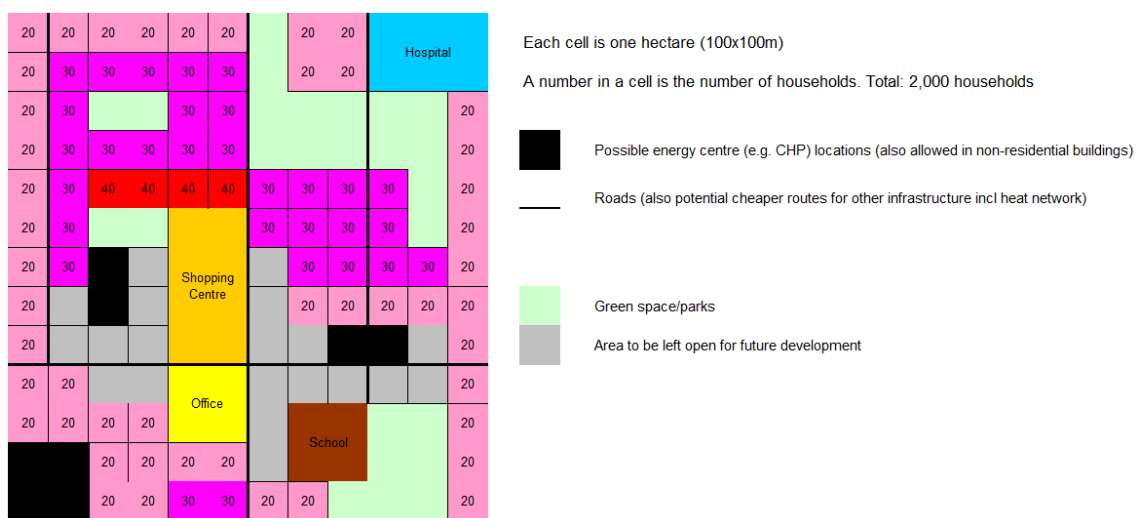
- Consideration of multiple alternatives for the whole urban energy system and comparison of relevant decision-making criteria
- Cost estimates (CAPEX as well as OPEX) for the different scenarios
- Annual operational CO<sub>2</sub> emissions (and other environmental impact, if appropriate)
- A stakeholder assessment and description of the impact on the main stakeholders, and selection of main problem owner (i.e. who takes responsibility for implementing the project... the party you are advising with your analysis and report)
- A discussion on business plan(s) covering briefly the ownership, design, financing, building and operation of your preferred energy system

The **final deliverable of the project** is a report that describes your approach, results and any conclusions you can draw from this work and emphasises the rationale for the chosen methodology and solutions. There is not a single right answer in this project, so a clear justification and reflection on the project outcomes is an important part of this deliverable.

The **maximum length** is 8 pages for the main body, plus title page, abstract/executive summary, references and an appendix, if appropriate. If you do use an appendix (e.g. with detailed calculations, tables with additional results, screenshots, or elaborate model descriptions) please make sure to refer to the content in the body of the report highlighting relevant findings and ensuring the main message is within the report. You should use the model definition prepared while designing the model (i.e. objective function, variables, parameters, main equations) as an appendix to your report to document your methodology. Also, please include an executive summary (which does not count towards the page limit) briefly summarising the report especially the main recommendation.

An **example report structure** is as follows, though you are free to choose a different suitable outline if preferred:

1. Executive Summary
2. Background and problem definition
3. Methodology and data collection
4. Results of baseline and sustainable alternative system designs
5. Analysis and business model description
6. Conclusions
7. Appendix (including model definition)



**Figure 1 – District layout**

We will be teaching you two different **optimisation tools**: AIMMS and GAMS Py. As you will see, you can express the same problem in both tools, though each has their own advantages. You only have to use one of these two options for your analysis. You can also use both tools, as well as any other software, if you see fit, though again using more software packages than just one does not necessarily mean a higher grade and you will not be penalised if you use just one analytical tool e.g. just AIMMS or just GAMS Py. Please note that for the baseline scenario no optimisation is required (there's nothing to decide, so there's no decision variable!) so you could just use your favourite software (e.g. Excel, Matlab, or R) for these calculations.

A brief note on **transport and other infrastructure** in the district. As said above, the focus in this project is on heat (including hot water) and electricity demand, so you do not have to consider (energy) demand from the transport sector, or infrastructure for drinking water, wastewater, recycling, etc. You will have a separate SEF module on sustainable transport, for example, and let's assume there is a public transport hub just on the edge of the new district and bus stops and high quality cycle lanes throughout. However, if you think it's relevant and might make your solution more effective, you are free to mention additional infrastructure where relevant. For example, if you think local electricity storage is important to balance demand and think that electric vehicle batteries can play a role, then vehicle to grid (V2G) could be part of that, and if you think energy from waste is relevant then mention it in this context. Do not let that make your project or your model more complex, but you are free to incorporate elements where this will make your solution easier to implement or more grounded.

Some cells in the city are candidates for a **potential energy centre**, if you think a more centralised solution serving several or even all buildings is an option, or any other relevant energy infrastructure. An energy centre can also be placed in one of the non-residential buildings (e.g. a CHP in the shopping mall which is oversized to also supply energy to some other buildings). As you will see on the layout plan, there are some cells to be left open for future development. Again, you can just consider those as empty cells with zero demand and that is absolutely fine and you should definitely start there, but if it helps your narrative you can also make proposals/assumptions for how that might develop in the future. You can also use future development as a justification for oversizing an energy centre to accommodate future demand if it leads to a better return of investment, for example. Would the overall system be more efficient if there was more high density housing in the area? Would a data centre in the proposed office building provide waste-heat which could be input for a heat network? Is there another kind of business you would like to attract to the area to help the electricity and heat system to become more sustainable? You have some flexibility to shape the future of the proposed development if it helps with the solutions you are exploring, or leave it out if it is not relevant. If there is something you would like to try regarding the design or assumptions and you are not sure if it is appropriate then just ask.

### 3 Group work and individual responsibilities

For this project you have been **assigned into groups** of 5 students, and please work in your assigned team. Together, the team will prepare one report following the guidelines presented above. Additionally, each student is expected to submit an individual report (max 750 words) outlining their contribution to the group work and a reflection on the findings and implications of the results. Please add name(s) and CID as well as your group number on the title page of all reports you submit, and use a suitable filename for your submissions which includes the group number for the group report and your surname for the individual report.

It may be efficient to (temporarily) assign **specific roles** to individual team members, so several tasks can be done in parallel. For example, a team could choose to have one person focus on data collection, one on the stakeholder assessment and decision criteria, two on setting up the optimisation model and one on exploring business models, and then change these roles as the project progresses. It should be stressed that this is a collaborative effort, and the best teams work closely together and communicate openly throughout the project, giving honest feedback and offer support. The earlier you start writing the report the better; so do not wait until after you have results from the optimisation model. If you use the draft report to store notes on literature, summaries of group discussions and presentation of early findings, you will have everything in one place to produce the final report. To prepare for your individual report, take a brief moment daily to reflect on your work and to make some notes on topics and examples you'd like to include, and keep working on this in parallel to the group work.

There are three **project surgeries** timetabled in your lecture plan. Please make the most of them to get feedback on your progress and help with the implementation of your optimisation model and to ask questions about the interpretation of the results and placing findings in a wider context. Before you start implementing a model, it is sensible to first design this on paper, just like we will do in the optimisation lecture and the AIMMS & GAMS Py tutorial sessions. Aim to have this done by the end of Monday 19<sup>th</sup> of January at the latest. Feel free to submit a short document with the model definition including objective function, main equations, variables etc for feedback. When you ask questions about the model and implementation, please have this description ready as that will help make clear what you are trying to do and why. This model definition should also be added to the appendix of your report to complement your methodology description, so keep it updated while you work on the model. **Please start small** and build on your model when you are satisfied you are getting relevant results.

### 4 Assessment

The module will be assessed by the group assignment over the two week period (80%) as well as your individual statement and reflection on this work (20%). The group assignment will be assessed based on the following criteria:

- **Problem definition** (15%) – Have the students critically examined the problem description, listed relevant stakeholders, identified potential analysis methods and clarified their working assumptions?
- **Methodology** (30%) – Is the chosen methodology sound, suitable for the objectives, well-executed and clearly described and justified?
- **Evidence of research** (20%) – Have the students sought knowledge outside the basic lecture notes? Has this been critically evaluated and sensibly used in their analysis?
- **Analysis** (25%) – Are the findings interpreted, critically examined, and combined in a meaningful way? Have the outcomes been placed in a wider socio-technical-economic context? Have the students explored and recommended a suitable business model and the potential implementation in the real world?
- **Presentation** (10%) – Is the final report well-presented? Are figures and tables clear? Have references been used appropriately?

Your individual report will be assessed based on the following criteria:

- **Reflection on own contribution** (40%) – Has the student clearly explained their own role in the project, reflected on their contributions to the team efforts, and on how well the group worked together? Have effective and specific examples been used in support?
- **Reflection on the chosen approach and outcomes** (60%) – Can the student identify strengths and weaknesses of the followed approach, and propose suggestions for further analysis? Has the student made good use of the lectures from this module when describing potential next steps?

As stated above, the page limit for the group report is 8 pages, plus frontmatter, references and an appendix if appropriate. Your individual report has a suggested length of 500 words and a maximum length of 750 words. The submission deadline for the group report (one per group) is 16:00 on Friday the 23<sup>rd</sup> of January 2026, and for the individual report (one per student) it is 18:00 on Friday the 23<sup>rd</sup> of January 2026.

## 5 Further reading

The book “Urban Energy Systems: An Integrated Approach” by Keirstead and Shah (Eds), Routledge, 2013, ISBN 0415529026, forms part of the reading material for this course and can be found as e-book as well as hardcopy through the Imperial College Library. Chapters 8 and 9 look at urban modelling and optimisation, and might be a useful starting point for your project. Additional material that could be useful can be found below, as well as in the slides presented in the lectures. Of course, looking up additional and alternative sources for your project where appropriate is highly recommended, and this list is merely meant as a starting point. Also include academic literature, where this could be relevant. Please clearly cite all sources used in your report.

- **The Future Homes and Buildings Standards** (Updated 11 December 2025)

<https://www.gov.uk/government/consultations/the-future-homes-and-buildings-standards-2023-consultation/the-future-homes-and-buildings-standards-2023-consultation#performance-requirements-for-new-buildings>

- **London Heat Map and London Heat Network Manual:**

<https://apps.london.gov.uk/heatmap/>

[https://www.london.gov.uk/sites/default/files/london\\_heat\\_map\\_manual\\_2014.pdf](https://www.london.gov.uk/sites/default/files/london_heat_map_manual_2014.pdf)

[https://www.london.gov.uk/sites/default/files/211004\\_lhnmii\\_master\\_final.pdf](https://www.london.gov.uk/sites/default/files/211004_lhnmii_master_final.pdf) (2021 updated)

- **Scotland heat map:**

<http://www.gov.scot/heatmap> (updated 2024)

- **Scotland's energy strategy and policies:**

<https://www.gov.scot/publications/scotlands-climate-change-plan-2026-2040>  
<https://www.gov.scot/groups/energy-standards-review-scottish-passivhaus-equivalent-working-group/>  
<https://www.gov.scot/publications/community-benefits-net-zero-energy-developments-analysis-responses-consultation-exercise/> (October 2025)  
<https://www.gov.scot/publications/heat-network-fund-application-guidance/> (September 2025)  
<https://www.gov.scot/publications/scottish-energy-strategy-future-energy-scotland-9781788515276/>  
<https://www.gov.scot/publications/annual-energy-statement-2020/>  
<https://www.gov.scot/collections/quarterly-energy-statistics-scotland/> (Q3 2025)  
<https://www.gov.scot/publications/community-benefits-net-zero-energy-developments-consultation/> (December 2024)  
<https://www.gov.scot/publications/local-energy-policy-statement/>  
<https://www.gov.scot/publications/heat-policy-statement-towards-decarbonising-heat-maximising-opportunities-scotland/>  
<https://www.gov.scot/publications/consultation-scottish-energy-strategy-future-energy-scotland-analysis-responses/>  
<https://www.gov.scot/publications/scotlands-energy-strategy-transition-plan-ministerial-statement/>  
<https://www.gov.scot/publications/heat-networks-delivery-plan-review-report-2024/>  
<https://www.scottishfuturestrust.org.uk/publications/documents/strategic-heat-network-development-route-map>

- **Community heating for planners and developers:**

<http://www.communityplanning.net/pub-film/pdf/CommunityHeating.pdf>

- **UK Government Energy Security Bill factsheet: Low-carbon heat scheme (September 2023)**

<https://www.gov.uk/government/publications/energy-security-bill-factsheets/energy-security-bill-factsheet-low-carbon-heat-scheme>

- **The Future of Heating: meeting the challenge (especially Chapter 3, and Tables 3: barriers and Table 4: policies): (quite old, but still interesting)**

[https://www.gov.uk/government/uploads/system/uploads/attachment\\_data/file/190149/16\\_04-DECC-The\\_Future\\_of\\_Heating\\_Accessible-10.pdf](https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/190149/16_04-DECC-The_Future_of_Heating_Accessible-10.pdf)

- **Assessing the future heating and cooling needs of the UK housing stock**

<https://assets.publishing.service.gov.uk/media/677d13c099c93b7286a39746/cs-now-assessing-future-heating-and-cooling.pdf> (2023)

- **The UK's Heat and buildings strategy (policy paper, 2023)**

<https://www.gov.uk/government/publications/heat-and-buildings-strategy>

- **Delivering London's energy future:**

<https://www.london.gov.uk/what-we-do/planning/london-plan/new-london-plan/london-plan-2021>  
[https://www.london.gov.uk/sites/default/files/london\\_plan\\_2021\\_sustainability\\_statement.pdf](https://www.london.gov.uk/sites/default/files/london_plan_2021_sustainability_statement.pdf)  
[https://www.london.gov.uk/sites/default/files/london\\_environment\\_strategy\\_0.pdf](https://www.london.gov.uk/sites/default/files/london_environment_strategy_0.pdf)  
<https://www.london.gov.uk/programmes-strategies/planning/planning-applications-and-decisions/pre-planning-application-meeting-service/energy-planning-guidance>

- **Analysis of the UK potential for CHP in 2030 (especially section 3 and annex 4):**

[https://www.gov.uk/government/uploads/system/uploads/attachment\\_data/file/191543/Projections\\_of\\_CHP\\_capacity\\_\\_use\\_to\\_2030\\_2204.pdf](https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/191543/Projections_of_CHP_capacity__use_to_2030_2204.pdf)

- **The U.S. EPA's CHP Catalog of CHP Technologies:**

<https://www.epa.gov/chp/chp-technologies> (updated March 2025)

- **The UK Potential for Community Heating with Combined Heat & Power:**

<https://www.gov.uk/government/publications/the-national-comprehensive-assessment-of-the-potential-for-combined-heat-and-power-and-district-heating-and-cooling-in-the-uk>

- **Energy in the health sector**

[https://www.england.nhs.uk/wp-content/uploads/2021/05/HTM\\_07-02\\_Part\\_A\\_FINAL.pdf](https://www.england.nhs.uk/wp-content/uploads/2021/05/HTM_07-02_Part_A_FINAL.pdf)  
<https://www.england.nhs.uk/greenernhs/a-net-zero-nhs/>  
<https://www.gov.scot/publications/nhs-scotland-climate-emergency-sustainability-strategy-2022-2026/>

- Operational data for the UK electricity national grid demand, generation and emissions:

<http://gridwatch.co.uk/>

<https://electricityproduction.uk/>

- Office for National Statistics (data on population, built environment, transport, etc):

<https://www.ons.gov.uk/>

<https://geoportal.statistics.gov.uk/>